Imitation Holds a Key to Evolution: Localized Repair and Verified Skill Acquisition for Adaptive Agents

Kai Hu · Hua Huang · Beijing International Studies Univ. · Chongqing Univ. of Posts and Telecom. · arXiv 2610.16125v1

HarnessRSI 68 ★★

evidence: retained

Paper

Code

1 What changes, what is kept

Symbolic controller, no neural model, 3 stages

Stages	select · resolve revision · normalize	**Actions**	2 / 3 / 3 opaque ids 18 programs in all
Feedback	violated path → producing stage	**Kept state**	evidence table per (revision, stage, goal)

1 Observe discrepancy
first full submission fails a stage contract; path → producing stage, request → local goal

2 Propose one local patch
candidates: acquired evidence, then fragments with same stage + effect, then any action

3 Small trial (≤ K)
run patched program on 2 order histories; keep the patch only if the target contract passes

4 Full-task audit
second submission vs an independent reference computation decides acceptance

5 Record evidence, gate reuse
trials add counts; proactive reuse needs 2 accepted tasks, q ≥ 0.7; frozen in test

later tasks

10 worlds · 96 acquisition tasks · 240 final tasks each
no gradient updates; the kept artifact is the evidence table

Every small trial adds success/failure counts and an incentive value, whether or not the full task is later accepted. Acquisition and evaluation are separate calls (state hashes).

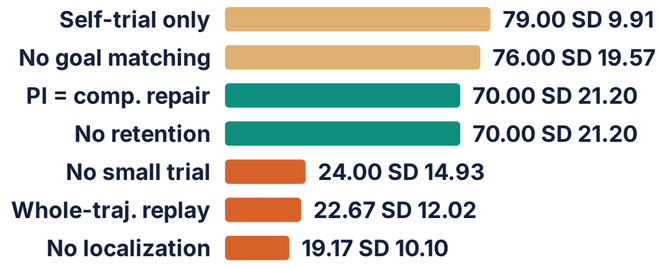
2 Headline: success (%) by controller, K = 3

Controller	Cold final	Learned first	Unseen C final
Whole-traj. replay	6.17	6.17	22.67
Full PI	**100.00**	**100.00**	**70.00**
No localization	9.50	100.00	19.17
No goal matching	92.50	100.00	76.00
No small trial	100.00	100.00	24.00
No retention	100.00	6.17	70.00
Comp. repair	100.00	100.00	70.00
Self-trial only	80.33	100.00	79.00

Success %, K = 3 (Table 1); cold = no acquired evidence; learned = after 96 atomic tasks. Comp. repair equals PI in every column.

3 Controls that decide if the gain is real

Unseen revision C, final success % (SD over 10 worlds)



Retention: first success after 96 tasks

With **100.00**, without **6.17**. On unseen C retention adds nothing (70.00 both).

Exploratory C stream: first success %

C tasks seen	0	6	12	24
PI first success	6.17	54.33	92.33	**100**
Self-trial first	6.17	37.33	91.17	**100**
No retention first	6.17	6.17	6.17	**6.17**

Declared after the primary results. Same frozen C tasks each time; stream = each of 12 combinations twice. Comp. repair follows PI.

Matched budget and the tie

Same interpreter, tasks and K for all. **Comp. repair** (no score threshold, no fragment-evidence ranking; shares some ordering) **matches PI** on success; trials, work, collateral reported for cold K = 3 only. Paper: no *unique* incentive advantage.

4 Where the gain is absent or small

- **Unseen C:** PI 70.00 vs self-trial 79.00 and no goal matching 76.00 — gaps smaller than the SDs (21.20, 9.91, 19.57), untested
- **K = 1:** PI 21.5 vs 24.0 without trials (no SD). K = 6: PI, comp. repair, self-trial all 100; self-trial fewer trials (2.46 vs 2.67)
- No small trial also 100.00 cold: 0.00 trials, work 58.82 vs 75.90
- Retention, familiar compositions: work 75.90 → 28.98 (interpreter proxy, acquisition not counted)
- Authors: one synthetic domain, supplied feedback; SDs across worlds, not task CIs; evidence table grows; no LLM baselines

Takeaway

In one synthetic domain, patching one stage at a time and reusing an action proactively only after two full-task confirmations improves later first attempts. An equally expressive control ties it; no LLM agent is tested.

Our read · HarnessRSI

Score 68 · ★★ · evidence level: retained

Reports: held-out ids, retention ablation, same K, 10 worlds, code.
Does not report: LLM setting, token cost, external agent baselines.